

Review of Machine and Deep Learning Techniques for Expression based Facial Emotion Recognition

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Abstract—Facial Emotion Recognition (FER) is a technique that learns features from raw images and achieve Emotions, is used to detect and analyze any person's emotional state from expressions based on facial cues. The ability to accurately recognize facial emotions must be improved to enhance virtual interpersonal and Human-robot interaction that can be beneficial in many applications, this can be achieved by leveraging Artificial Intelligence and computer vision techniques. This paper studied the advancements in the field of FER, that can influence performance and compared different Machine Learning(ML) and Deep Learning(DL) based techniques by focusing on several aspects: like Nature of dataset, the input representation for providing more standardized input data, and choice Network architectures for feature extraction and classification purposes. In essence, this paper serves as a comprehensive guide to FER, providing insights into its current landscape, challenges, and future directions.

Index Terms—Face Detection, Facial Emotion Recognition, Emotion Classification, Machine Learning, Deep Learning, Hybrid.

I. INTRODUCTION

The era of digital communication marked by the rise of digital devices and the internet, these devices like smartphones, tablets, computers, and CCTVs, are mostly equipped with cameras which create a vast amount of digital data like images or sequence of images. However, this has made it difficult and time-consuming for people to find and manage the important information from this huge amount of digital data. Facial emotion recognition is the method of extracting valuable emotion from digital data like Images and sequence of images (i.e. video streams) that contain facial images. As is widely known that, the face is the most expressive and communicative aspect of a human being because it may convey various emotions through facial expressions without saying a single word [1]. American psychologists Friesen and Ekman identified six primary emotions in the 20th century, namely surprise, happy, sad, anger, disgust, and fear which are common across different cultures [2]. FER is a technology that detects, analyzes, and interprets a person's emotions through facial expressions. The aim is to train a model which can correctly categorize a given input image of a face into one of several emotion categories.

In FER rather than pre-processing of image, Feature extraction and classification techniques are also major parts for

which different kinds of models can be used, are covered in Section III. A) Traditional ML-based models like Support Vector Machines(SVM) [3], k-Nearest Neighbours(k-NN) [4], Random Forest(RF), etc. B) DL-based models like Convolutional Neural Network(CNN) [5], Recurrent Neural Network(RNN) [6], Long Short-Term memory(LSTM) Network [7], etc. C) Hybrid model, by combining different models.

FER can be beneficial in many AI applications, enabling machines to better understand and interact with humans [1]. It has the ability to improve understanding of emotions and their impact on behavior, and it can have practical applications in the fields of marketing [8], entertainment, shopping, e-commerce, education [9], restaurant, and health sector feedback [10]. e.g., In education technology to gauge student engagement and adapt learning materials to individual emotional needs, resulting in more effective learning experiences [11]. In psychology to identify signs of depression or anxiety in a patient and in Healthcare for understanding patient's emotions, which can aid in early invention and treatment for providing better care [12].

This study aims to present a systematic review of the different prior studies and Literature to gain deep insight into the FER. The goal is to support researchers and academics to discover the research outcomes and future directions of FER. This paper is arranged in the following sequence: Section II thoroughly discusses the overview of FER, where it discusses the steps to perform FER, like dataset acquisition, different image pre-processing steps, and also list out often used with relevant details, and further feature extraction and classification steps. Section III includes a literature review that provides insights from various FER studies. Section IV discusses the several challenges identified in the literature surveyed, along with directions for future work. Section V represents the study's conclusion.

II. OVERVIEW OF FACIAL EMOTION RECOGNITION

Facial emotion recognition [12] is a technology for detecting and analyzing facial expressions [7]. It involves an image of someone's face and analyzing it for determine the same person's emotional state based on facial expressions such as eye dilation, shape of the mouth, and eyebrow movement. FER

aims to create an automated emotion recognition system that can accurately detect and categorize emotions from the input image, by leveraging Computer vision, and Artificial Intelligence techniques to create a framework that can identify the specific facial expressions associated with different emotions like surprise, happiness, sadness, anger, disgust, fear, neutral [13], etc. It involves a series of steps that shows in Fig. 1, Here's a general methodology that can be followed.



Fig. 1. Basic steps to perform FER

A. Dataset Acquisition

FER are problem-specific, and these problems are basically related to which type of image is used as an input image. The accuracy of a FER can be significantly biased towards the dataset it was trained on. Various datasets are available to address different challenges [11]. For instance, the FER2013 dataset is commonly used to tackle real-world problems such as occlusion and varying lighting conditions [9]. The JAFFE dataset, on the other hand, is highly balanced and is often used when the goal is about to achieve high accuracy with less computational time. The type of images in the dataset, whether they are spontaneous or posed, also plays a significant role in terms of the performance of FER system. Therefore, the selection of an appropriate dataset that aligns with the specific requirements [8]. Here Table I represents some datasets that are often used.

TABLE I
FACIAL EXPRESSION DATABASES WITH RELEVANT CHARACTERISTICS

Name of Dataset	No. of Subjects	No. of Emotion Classes	Name of Emotion Classes	Nature of Dataset	Colour of Images	No. of Images
JAFFE [1]	10	7(6Basic + Nu)	Su, Ha, An, Sa, Fe, Di + Nu	P	Grayscale	213
CK+ [7], [10]	123	7(6Basic + Co)	Su, Ha, An, Sa, Fe, Di + Co	P & S	Grayscale + RGB	593 (Images+ Sequence of images)
FER2013 [2], [14]	NA	7(6Basic + Nu)	Su, Ha, An, Sa, Fe, Di + Nu	S	Grayscale	35,887
RAF-DB [4], [15]	NA	7(6Basic + Nu)	Su, Ha, An, Sa, Fe, Di + Nu	P & S	RGB	29,672
Affect-Net [7], [16]	NA	8(6Basic + Co, Nu)	Su, Ha, An, Sa, Fe, Di + Co, Nu	P	Grayscale + RGB	0.4M
KDEF [15], [17]	70	7(6Basic+ Nu)	Su, Ha, An, Sa, Fe, Di + Nu	P	RGB	4,900

(Su: Surprise, Ha: Happiness, Nu: Neutral, An: Angry, Sa: Sadness, Fe: Fear, Di: Disgust, Co: Contempt, P: Posed, S: Spontaneous)

B. Image Pre-Processing

Pre-processing of an image is used for improving the quality of an image before applying any feature extraction technique. It can reduce noise and unwanted distortions in the input image, making it easier for further feature extraction

process. Images can pre-processed in the following steps and corresponding techniques listed in Table II:

1) Resizing of images can be used, making it more suitable for model training and inference. 2) For finding the face in the images, Landmark/face detection [12] methods are used. 3) For adjusting the brightness and contrast of the images, Light intensity Normalization methods are used which can enhance the images, and make them more clear and visible [12]. And 4) For aligning the faces to a standard position, Posture Normalization methods are used [2] [12]. After these steps, 5) Data Augmentation [8] [5] is used for generating new training image samples to increase the amount of the dataset, by applying some transformations such as flipping, rotating, shifting, etc. to the original images. It prevents models from overfitting.

TABLE II
IMAGE PRE-PROCESSING TECHNIQUES

Purpose	Image Pre-processing Techniques
Image Resize	Resize, Crop, Orient [8]
Face/Landmark Detection	Viola-Jones [1] [18] Dlib [19]
Light Intensity Normalization	Histogram Equalization [12] [20] Background Subtraction [21]
Posture Normalization	Frontalization [10] Face Alignment [20] Face Normalization [8] [22]
Data Augmentation [5]	Geometric transformation, Color space transformation

Image pre-processing can have many advantages, such as:

- It can increase the model performance and accuracy by providing more consistent and standardized input data.
- It can minimize the model's training time and increase the model inference speed by reducing the size and complexity of the image.

C. Feature Extraction

In FER task Feature extraction [10] is most crucial part. It is the process of extracting relevant and discriminative information from the input data without losing important or relevant information. It refers process of converting raw data into numerical features which can used for further analysis or learning. Feature extraction aims to reduce features in such a way that information remains the same or there should be negligible changes that do not affect the final outcome.

Feature extraction is important because it helps to dimensionality reduction of the data by removing Irrelevant and Redundant features, Removing noise [23] which can prevent overfitting and reduce computational cost also provides a better understanding of data that can further improve the efficiency and model's generalization performance of the FER.

D. Classification of Emotion

For FER task, next step is to categorize the emotions on the basis of extracted features using classifiers [22]. The classifier

is trained on labeled data, where each image is associated with an emotion class. The final output of classifier is a prediction for each image, indicating the most likely emotion based on the features it has learned during training. Fig. 2 represents classified Emotion with respect to input image.



Fig. 2. Classified Emotion with respect to input image [17]

The effectiveness of the emotion classification depends on the extracted feature's quality and the ability of the classifier to learn from those features.

III. LITERATURE REVIEW

In case of FER feature extraction and Classification are crucial parts for which different kind of models can be possible, for more refer Table III

A. Traditional Machine Learning Based Approach

FER was primarily achieved using traditional machine learning-based models such as the SVM [10], k-NN, RF, etc. These models relied on handcrafted face landmark-based geometry and appearance features [9] [23]. Techniques like Histogram of Oriented Gradients(HoG) [18], Gradient feature mapping, Eigenvectors, Gabor filters [27], Eigen Faces, Linear Discriminant Analysis(LDA) [4], Principal Component Analysis(PCA) [10], Local Binary Patterns(LBP) [10] [18], and so on were used to extract features, refer Fig. 3. However, the manual extraction of these features could lead to an error rate.

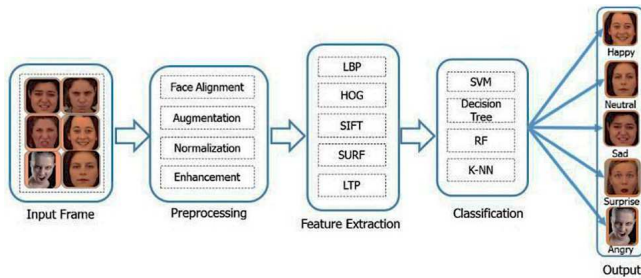


Fig. 3. Traditional ML-Based FER [28]

Dino et al. [3] present an automatic FER system which uses the Viola-Jones face landmark detection algorithm, HOG based feature extraction, and PCA method to reduce the dimensionality while retaining 90% of the original information. Three machine learning classifiers, namely SVM, K-NN, and MLP, were implemented for classification of facial emotions, and their effectiveness was compared. CK+ Dataset used to validate the results. The outcome rendered SVM as most accurate classifier with a recognition rate of 93.53%, followed by MLP with 82.97%, and KNN with 79.97%.

Adil et al. [27] presents a new method for recognizing facial expressions using ML techniques. As per their method,

they first detect the face landmark using the Haar cascade classifier, then extract the Regions of Interest(ROIs) from the face, which typically include the mouth region, left and right eyes, and eyebrows. Then, they apply the Gabor filter to these ROIs to extract features. After obtaining the feature vector, they reduce its size using PCA while maintaining 95% of the original information. Finally, the reduced feature vector is classified into one of seven emotions using the SVM classifier. Experiments on CK+ and JAFFE datasets led to recognition rates of 92.19% and 95.11% respectively.

Vatcharaphrueksadee et al. [23] discuss a framework for categorizing facial emotion for both cartoon-animated and human faces. This framework employs a combination of Appearance and Geometric based [9] feature extraction from facial images obtained from two datasets, the FER-2D Dataset and the CK+. Face detection is made via Multi-task Cascaded Convolutional Networks. Feature extraction involves the use of a HOG for appearance features, and a method by Youssif and Asker for geometric features extraction from different regions of a face such as eyes with eyebrows, nose, and mouth. To manage the high dimensionality of the data, PCA is applied to the appearance features. The final emotion classification is performed using a SVM, recognizing six emotions: joy, surprise, contempt, anger, sadness, and fear. And The recognition accuracy of the proposed solution is 92.63%.

B. Deep Learning Based Approach

With the Deep learning models, the landscape of facial emotion recognition has changed significantly. Unlike traditional methods, DL models have the ability to learn these features and extract them automatically during the training phase [10]. This automatic feature learning capability eliminates the requirement of manual handcrafted feature extraction, reducing the chance of error rate. Fig. 4 illustrates the typical Convolutional Neural Network Architecture which can be used for feature extraction and classification of FER task.

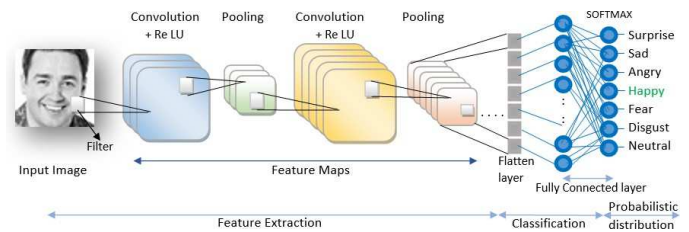


Fig. 4. Architecture of Convolutional Neural Network

- **Input Layer:** This layer receives the raw data (image) and represents it as a grid of pixel values.
- **Convolutional Layer [22]:** Moves a filter over the image to extract features, and further ReLU Activation Function is applied to introduce non-linearity, resulting in a convolved feature map [5].

TABLE III
A DETAILED REVIEW OF PRIOR LITERATURE ASSOCIATED WITH THE FER TASK

Author	Dataset	No.Of Classes	Image Pre-Processing Techniques	Model	Feature Extraction Technique	Classification Technique	No. Of Epochs	Optimizer	Accuracy
[14]	CK+	7	Data Augmentation	Deep Learning (CNN)	VGG13	SOFTMAX	200	Adam	99.18%
	JAFPE	7	Data Augmentation				200		98.75%
	KDEF	7	Data Augmentation + Grayscale				200		91%.
	FER2013	7	NA				500		74.17%,
[16]	FER+	8	Data Augmentation	Hybrid (CNN-SVM)	VGG-Face	SVM	400	Not Mentioned	87.76%
	FER2013	7			VGG-f		600		75.42%
	AffectNet	7			VGG-13		200		63.31%
		8							
[13]	FER2013	7	Haar Features	Deep Learning (13 Layer CUSTOM CNN)	10 Convolutional + max pooling) Layers	(1 Flatten, 1 Dense, 1 Output (Softmax)) Layers	200	Not Mentioned	67%
[7]	CK+	6	Haar Features +Grayscale Conversion	Hybrid (CNN-LSTM)	CNN	LSTM	100	Adam	90.51%
[24]	FER2013	6	Dividing dataset classes P-CNN(Ha,Sa) S-CNN (Su,An,Di,Fe)	Hybrid CNN	P-CNN (3 Convolutional +MaxPool)	P-CNN (2 Fully connected + 1Fully connected with Softmax)	20	Adam	97.07%
	JAFPE				S-CNN (5 Convolutional + Maxpool)	S-CNN (2 Fully connected with 2 Dropout + 1 Fully connected with Softmax)			94.12%
[19]	CK+	8	DLIB	Hybrid (GA-SVM)	Genetic Algorithm (GA)	SVM	300	Not Mentioned	95.85%
	MUG	7							97.59%
[25]	FER2013	7	Image resize + Grayscale	Deep Learning (CNN)	AlexNet	SOFTMAX	50	RAdam	70.52%
	CK+	7							99.4%
[4]	CK+	7	Haar Features + Grayscale Conversion	Machine Learning (Gabor-k-NN)	Gabor filter	k-NN	Not Mentioned	Not Mentioned	99.75%
	RaFD								94.60%
	KDEF								86.02%
[17]	FER2013	7	Data Augmentation + Resizing by Bi-interpolation	Custom CNN	4 Convolutional + Maxpool	1 Flatten + 1 Fully Connected +1Fully connected with softmax	Not Mentioned	Adam	78.9%
	JAFPE								96.7%
	CK+								97.8%
	KDEF								82.5%
	RAF								81.68%
[8]	FER2013	7	Haar Features, Crop and Normalization	Hybrid (F-CNN-SVM)	F-CNN (fuzzy C-means clustering(FCM) +CNN)	SVM	Not Mentioned	Not Mentioned	83.86%
[26]	SFEW2.0	7	Face Alignment	Deep Learning (CNN)	VGG-16,	SOFTMAX	Not Mentioned	Not Mentioned	48.9%
	Oulu- CASIA	6			Resnet-50				54.8%
	EmotioNet	7(Basic)+ 9(Comp ound)				SIGMOD (For Action Units)			87.8%
									89.0%

- Pooling Layer: Generally uses max-pooling to shrink the feature map, enhancing dominant features and reducing computational complexity [2].
- Flatten Layer: Flattens the extracted numeric multi-dimensional array into a 1-D vector.
- Fully Connected Layer(s): Learns high-level features and patterns from the flattened vector.
- Output Layer: This is the last layer where the number of neurons is equal to the emotion classes to be classified. The softmax activation function [2] maps the vector into a probability distribution of emotion classes, and based

on that the highest probability of emotion class becomes the final classified emotion.

Nwosu et al. [1] describe a FER system based on a two-channel Deep CNN utilizing facial parts such as eyes and mouth as inputs. They emphasize the efficiency of this method, noting it reduces training time and cost and enhances recognition accuracy. The system uses the Viola-Jones face detection framework for identifying and extracting facial features from images. The extracted parts are downsized to reduce the quantity of data the network has to process, thus making training quicker. The authors tested their system on CK+ and

JAFFE datasets. The system achieved 95.72% and 97.71% recognition accuracy, respectively.

Mehendale [21] introduces a novel technique called FER using CNN(FERC). FERC employs two stages of CNN: one for background removal from the image and another for extracting facial features. Expression vector (EV) is used for further identifies five common facial expressions. FERC Trained on a dataset of 10,000 images from 154 individuals and extensive testing on 750,000 images using various datasets, achieved an impressive 96% accuracy despite acknowledging challenges like computational requirements and handling facial hair. The author foresees FERC as a foundational tool for emotion-based applications, envisioning its use in lie detection, mood-based learning environments, and diverse applications powered by emotion recognition technology.

Borgalli & Surve [15] propose a DL-based framework using CNN for FER. The study utilizes both controlled lab, and wild environments datasets. The datasets are RAF-DB, KDEF, RAFD, AM-FED+, and SFEW. Using these datasets, they train standard and custom CNN architectures i.e., InceptionV2 and VGG to recognize basic six emotions, along with neutral and contempt. The model's performance, depicted using confusion matrices, varies across datasets with accuracy ranging from 41% to 100%.

Khajuria et al. [29] presents a technique for recognizing emotions through facial expressions using a CNN and VGG-16 model. The authors propose a DL model that is pattern-based to achieve higher accuracy. The process involves recognizing the position and motion of facial muscles, pre-processing data, extracting features, and classifying facial emotions. The authors note that while machine learning algorithms have been used by researchers to extract facial emotions, they did not achieve optimal accuracy. To address this, the authors propose a deep learning model that is pattern-based to achieve higher accuracy. The dataset used in this work is FER2013. The result of proposed model achieves an accuracy of 91%.

C. Hybrid Model-Based Approach

Hybrid models can be created by combining different models for feature extraction and classification, allowing to take advantage of the benefits of each model. Liu et al. [30] propose a new model for fused SVM and Sobel edge detection based on an improved CNN. The authors use the VGG11 model to extract facial expression features and change softmax activation function with L2-SVM, for facial expression classification. To assess their approach, the authors perform experiments on the CK+ and JAFFE datasets. These experiments indicated that the substitution of softmax activation function to L2-SVM enhanced accuracy by 3.02% and after including pre-processing of image using Sobel edge detection accuracy is increased by 3.71% on both the CK+ and JAFFE datasets.

Li et al. [7] presents a development of an emotion recognition system for a humanoid robot, aimed at improving human-robot interaction. For analyzing facial expressions system uses both a CNN and LSTM network. The CNN identifies visual features from static facial images, whereas

the LSTM determines the connection between the alteration of facial expressions from sequences of image and associated emotions with them. The study leverages transfer learning to enhance the system's recognition accuracy, by transferring acquired knowledge from a large collection of static data to a smaller collection of dynamic data. This method is validated against other models using leave-one-out cross-validation. Six primary emotions are recognized by the system. Harley (Robot) captures users' facial expressions with its camera and, using the proposed model, recognizes the emotion and replies appropriately. The detection archives increased success rate from 58.62 to 90.51%.

IV. CHALLENGES AND FUTURE DIRECTIONS

FER task is problem-specific, by going through literature, observed some open issues, limitations, and challenges that need to be addressed in future work.

A. Nature of Dataset

The FER task is biased towards the dataset [22], and the nature of dataset plays a crucial role, as it affects the performance of model and the ability to generalize across various cultures and individuals, age [10], sex, ethnicity, and also Nature of datasets like Posed (lab-controlled) and Spontaneous (real-world scenario based) [12]. To address this acquisition of more diverse and balanced databases which include a huge range of facial expressions from various demographic groups is used to reduce biasness and improve model generalization.

B. Overfitting

Overfitting can introduced when a model is too complex and learns too well from the learning (training data), including irrelevant outliers and noise. As result, it outperforms the learning data but in case of unseen data it performs poorly because it fails to generalize the underlying patterns. To address this, sufficient regularization techniques can be used because regularization further adds loss function to a penalty, which discourages overly complex models that can lead to prevent overfitting. Techniques like L1/L2 regularization [9], dropout [5], weight decay, early stopping, etc. are used to get a smooth curve between the training and testing phases.

- **Class Imbalance:** Mostly class imbalance issue occurs in case of real-world based datasets like FER2013, CK+, RAF-DB, etc. Class imbalance means if certain classes are minority compared to the majority classes and due to this, the model performs poorly on minority classes which also causes the overfitting problem [22], to avoid this methods like data augmentation, weighted loss function, etc. can be used. Which further helps to balance the classes by creating more data.

C. Model Essembling

To improve the model's performance in terms of computational time, resources, and accuracy different datasets or different models (Hybrid) can be essemble together [22]. Face landmark detection algorithm with different image pre-processing

techniques, followed by CNN Transfer learning [5], and fine tuning [9] can further help for better learning (training) of emotion based on extracted feature and classification by adjusting the Hyperparameters of models, which can be used as a better choice for future FER task.

V. CONCLUSION

This review summarises various Machine learning-Deep learning based approaches that are used for Facial Emotion Recognition (FER) task. Deep learning models have shown superior outcomes in comparison to traditional ML models, and in case of Hybrid model, where different combinations of model are used for different purposes to take advantage of these selected models. This study surveyed the recent developments in FER, and discussed the various factors that influence the performance of FER, like the choice of datasets, the pre-processing steps of image, and the techniques that are used for extraction of features and further classification purposes, and also compared different approaches for each factor. This paper provides a comprehensive guide to FER, covering its evolution, current landscape, challenges, and applications. CNN-based transfer learning is mostly used to enhance the performance ability of the model in terms of better recognition rate and also saves computational resources. The discussed methods demonstrate that deep learning models can revolutionize the field of FER by providing automated feature extraction and learning in the training phase, timely, and accurate emotion analysis. In future Hybrid model can be used for betterment in the field of FER.

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